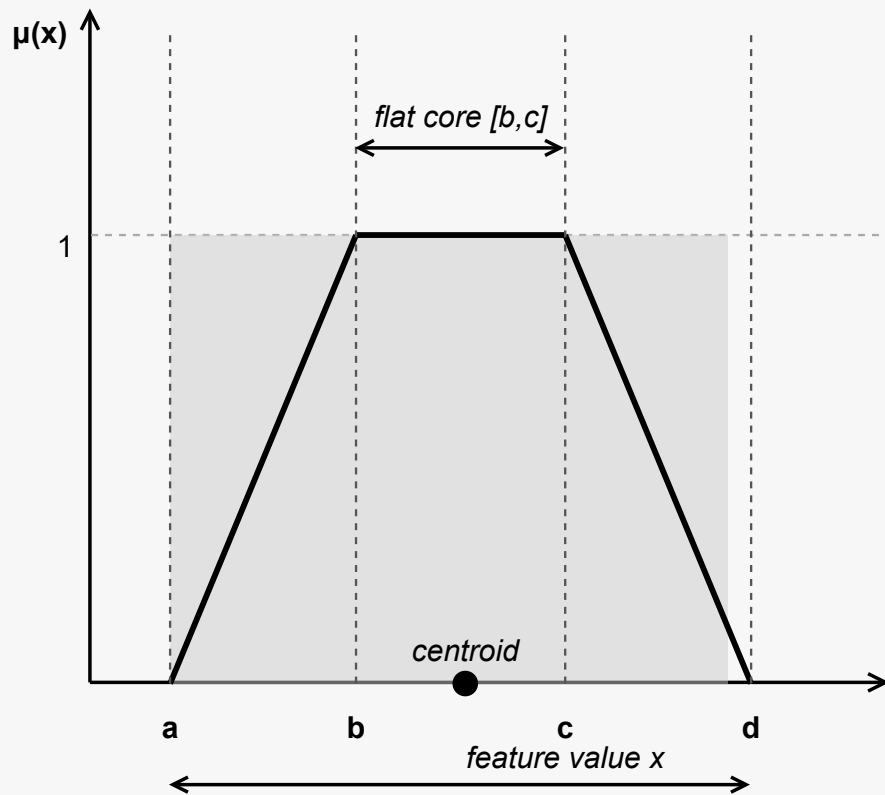


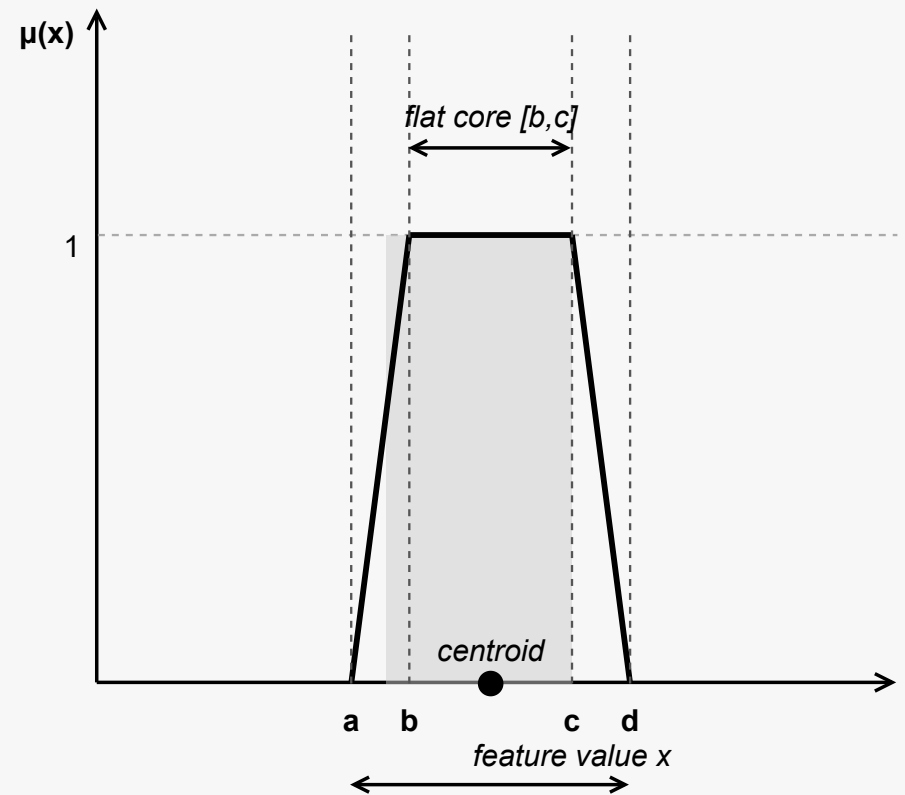
### High Uncertainty (wide interval)



**support =  $d - a$  (large)**

Large support → **low sharpness**  
→ diffuse soft routing at tree node

### Low Uncertainty (narrow interval)



**support =  $d - a$  (small)**

Small support → **high sharpness**  
→ near-deterministic routing at tree node

$\eta$  (proportion factor) controls the ratio of the flat core  $[b, c]$  to total support  $[a, d]$ . Observed entries degenerate to a point:  $a = b = c = d$ .

## Phase 2 — Proportional Fuzzy Transformation